

Pharmacokinetic Factors Associated With Early Meropenem Target Attainment in Pediatric Severe Sepsis

OBJECTIVES: To determine the frequency of early meropenem concentration target attainment (TA) in critically ill children with severe sepsis; to explore clinical, therapeutic, and pharmacokinetic factors associated with TA; and to assess how fluid resuscitation and volume status relate to early TA.

DESIGN: Retrospective analysis of prospective observational cohort study.

SETTING: PICU in a single academic quaternary care children's hospital.

PATIENTS: Twenty-nine patients starting meropenem for severe sepsis (characterized as need for positive pressure ventilation, vasopressors, or ≥ 40 mL/kg bolused fluid), of which 17 were newly escalated to PICU level care.

INTERVENTIONS: None.

MEASUREMENTS AND MAIN RESULTS: Concentration-time profiles were analyzed using modeling software employing opportunistic sampling, Bayesian estimation, and a population pharmacokinetic model. Time above four times minimum inhibitory concentration ($T > 4 \times \text{MIC}$), using the susceptibility breakpoint of $1 \mu\text{g/mL}$, was determined for each patient over the first 24 hours of meropenem therapy, as well as individual clearance and volume of distribution (Vd) estimates. Twenty-one of 29 patients met a target of $40\% T > \text{MIC}$ $4 \mu\text{g/mL}$. Reaching TA, vs. not, was associated with lower meropenem clearance. We failed to identify a difference in Vd or an association between the TA group and age, weight, creatinine-based estimated glomerular filtration rate (eGFR), or the amount of fluid administered. eGFR was, however, negatively correlated with overall $T > \text{MIC}$.

CONCLUSIONS: Eight of 29 pediatric patients with early severe sepsis did not meet the selected TA threshold within the first 24 hours of meropenem therapy. Higher clearance was associated with failure to meet targets. Identifying patients likely to have higher meropenem clearance could help with dosing regimens.

KEYWORDS: children; critically ill; meropenem; pharmacokinetics; sepsis

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Pediatric sepsis—defined as “infection with life-threatening organ dysfunction” by the 2024 Phoenix International Consensus Criteria (historically “severe sepsis” by the 2005 International Pediatric Sepsis Consensus Conference definitions)—is a common cause of morbidity in the PICU and carries a 7% risk of mortality in even high resource settings (1, 2). Early appropriate antibiotics is evidence-based treatment (3), and cohort studies from 1989 to 2016 show that timing of appropriate therapy affects outcomes (4–7).

Beta-lactam antibiotics—worldwide the most commonly used class of antibiotic in hospitalized children in 2012 (8)—have a bactericidal effect dependent on percentage of time that concentrations remain above the minimum inhibitory concentrations of target bacteria ($\%T > \text{MIC}$) (9). Meropenem can be

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RESEARCH IN CONTEXT

- The key therapy for severe sepsis is early initiation of appropriate antibiotic therapy.
- Beta-lactam efficacy relates to time above target bacterial minimum inhibitory concentration ($T > MIC$).
- Critical illness changes antibiotic pharmacokinetics and impacts $T > MIC$. This study seeks to identify factors associated with early meropenem target attainment in pediatric severe sepsis.

empiric treatment for presumed severe Gram-negative infections in children with history of antimicrobial resistant infections. Suggested meropenem pharmacodynamic targets vary in the literature, ranging from 40% $T > 1 \times MIC$ to 100% $T > 4 \times MIC$, depending on whether studies were in vitro or clinical (9–12). The landmark Defining Antibiotic Levels in ICU Patients (DALI) cohort study from 2014 showed that many adult patients, in retrospect, do not meet stringent $T > MIC$ targets for beta-lactams (13), and patients failing to meet targets may have less likelihood of good outcomes. Studies assessing meropenem target attainment (TA) in critically ill children via trough concentrations (14) or population pharmacokinetic (PopPK) models and Monte Carlo simulations also suggest that many do not meet stringent $T > MIC$ targets with standard dosing (15–19).

The concentration-time profiles used to calculate $T > MIC$ are based on pharmacokinetic parameters like volume of distribution (V_d) and clearance, which are patient- and pathology-specific. The pathophysiologic changes inherent to early dynamic critical illness alter pharmacokinetic parameters (20). V_d is affected by dehydration or capillary leak. Clearance may be increased by augmented renal clearance (21, 22) or decreased by acute kidney or hepatic injury. Co-administered therapies—namely fluid resuscitation—may also impact pharmacokinetic parameters. Although antibiotics and fluids are often used concurrently in early sepsis (3), the associations between volume resuscitation and overall fluid status with antibiotic TA are unknown.

Because of dynamic physiologic derangements, a 2020 European Infection and Intensive Care Societies position article jointly suggested routine monitoring of beta-lactam concentrations in critically ill adults (23). However, a systematic review of pharmacokinetics and TA of antibiotics in critically ill children (through 2017) showed that “studies frequently fail to provide dosing advice... and [there are] gaps in current knowledge” (24). Therefore, we sought to determine: 1) how often critically ill children with severe sepsis obtain target concentrations within the first 24 hours of meropenem therapy and describe their pharmacokinetic parameters and 2) assess how acute fluid resuscitation and subsequent volume status relate to TA. We hypothesized that many children with severe sepsis fail to meet goal concentration thresholds and that higher clearance and V_d occur in patients failing to meet $T > MIC$ goals compared with those meeting targets. Further, we hypothesized that patients receiving higher volumes of resuscitative fluid would have higher V_d and thus potentially lower % $T > MIC$.

MATERIALS AND METHODS

This study was a secondary, retrospective analysis of a curated dataset from a single-center prospective cohort study from 2018 to 2021 in the PICU at Cincinnati Children’s Hospital Medical Center. The parent study was approved by the Institutional Review Board (IRB) on September 2, 2018 (“Pharmacokinetics of β -lactams in Critically Ill Pediatric Patients during Different Stages of Sepsis,” No. 2018-3245), and used opportunistic sampling to study the pharmacokinetics of beta-lactams in critically ill children (ceftriaxone, cefepime, piperacillin/tazobactam, or meropenem) (25, 26). The current secondary analysis had an IRB-approved waiver of consent. Procedures were followed in accordance with the ethical standards of the responsible institutional committee on human experimentation and with the Helsinki Declaration of 1975.

For this study, we included patients in the cohort if they started meropenem for severe sepsis. We defined severe sepsis as treatment with antibiotics for greater than or equal to 7 days (or until death, if < 7 d) for presumed infection AND requirement of greater than or equal to 1 of the following: vasopressors within 24 hours of first meropenem dose, positive pressure

ventilation (invasive or noninvasive) above baseline within 24 hours of first dose, or greater than or equal to 40 mL/kg (or ≥ 2 L if weight was ≥ 50 kg) of bolus IV fluid within 12 hours of first dose. Finally, patients needed greater than or equal to 1 meropenem concentration measured via opportunistic sampling (described below) within the first 24 hours of therapy.

We excluded patients from the study cohort if they were not newly starting meropenem in the PICU (i.e., received ≥ 2 doses prior to PICU transfer). Patients were not excluded if they received other antimicrobials. Concurrent antimicrobials given on the calendar days including the first 48 hours of meropenem are shown in **Table S1** (<http://links.lww.com/PCC/C541>). Patients were also excluded from the cohort if they had had a surgical procedure or required molecular adsorbent recirculating systems or extracorporeal membrane oxygenation.

Target Attainment Definition

For most children with early severe sepsis, the target organism is unknown and antibiotic choice is made empirically. In our hospital (2018–2021), meropenem was often used to treat *Enterobacter* as the empiric target. For this study, TA was assessed using a goal concentration greater than $4 \times \text{MIC}$ of 1 $\mu\text{g/mL}$ (i.e., $> 4 \mu\text{g/mL}$) for 40% of the first 24 hours of therapy. One $\mu\text{g/mL}$ reflects the sensitivity breakpoint for *Enterobacter* per 2023 Clinical Laboratory Safety Institute (27).

Data Collection/Curation

Patient characteristics, medical history, PICU-admission-to-meropenem-initiation timing, laboratory values, concomitant therapies, and fluid balances related to the episode of severe sepsis were obtained from the electronic medical record during 2018–2022. Data were curated and stored in a Research Electronic Data Capture database (28). Outcomes were saved and included death by 28 days or the presence of greater than or equal to 2 organ system failures at 7 days.

At the time of the parent study, opportunistic samples of blood (i.e., remaining from studies ordered by the clinical team for patient care) were stored at 4°C (29), centrifuged to liberate plasma within 7 days of collection and saved for later measurement of total meropenem concentrations (25) via high performance

liquid chromatography (HPLC). HPLC was performed between January 2019 and December 2021. Samples obtained during the 30 minutes following meropenem infusion start were excluded. This methodology was previously validated by our group (25). Total concentrations (rather than free) were used given low protein-binding reported in the literature (30). The lower limit of quantification (LLQ) was 0.5 $\mu\text{g/mL}$. Concentrations less than 0.5 $\mu\text{g/mL}$ were imputed as half LLQ, that is, 0.25 $\mu\text{g/mL}$.

Pharmacokinetic Analysis

Drug concentrations were analyzed over time with the precision dosing software MwPharm++ (Mediware, Prague, Czech Republic) (31) employing a Bayesian estimation technique to determine concentration-time profiles based on a two-compartment meropenem PopPK model in critically ill children (18). This technique allowed for estimation of pharmacokinetic parameters and profile before steady state. The model by Saito et al (18) was selected because it was developed for critically ill children and included patients on continuous renal replacement therapy (CRRT). Based on the reported assay error from the clinical laboratory, assay error in MwPharm++ was set to 0.1 $\mu\text{g/mL}$.

For each patient, demographic data, all concentrations and meropenem doses within the first 24 hours and the next concentration or dose after 24 hours, were used to inform the fitted concentration-time profile analysis completed in April 2024. All patients received intermittent dosing of meropenem (dosing frequency shown in **Table 1**), and infusions were assumed to run over 30 minutes. Data required by the reference PopPK model include daily weight and estimated glomerular filtration rate (eGFR): bedside Schwartz equation ($k = 0.413$) was used for patients younger than 18 years old, and Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) formula was used for patients 18 years old or older (32, 33). Additional model covariates included CRRT requirement and a daily systemic inflammatory response syndrome (SIRS) score (assessed using the most abnormal vitals or laboratory values for the day). An example concentration-time profile generated is shown in **Figure 1**. These profiles were used to determine %T $>$ MIC across the first 24 hours. Vd and clearance were estimated for each patient; clearance was allometrically scaled to 70 kg and normalized to each patient's eGFR and SIRS score.

TABLE 1.

Patient (*n* = 29) Ages, Weights, Original PICU Admission Diagnoses, Starting Dosing Regimens, Culture Data, and 24-Hour Percent Time Above Empiric 4 µg/mL (> 4x Minimum Inhibitory Concentration of 1 µg/mL)

Patient Number	Age (yr)	Weight (kg)	PICU Admission Diagnosis	Day 1 Estimated Glomerular Filtration Rate ^a (mL/min/1.73 m ²)	Meropenem Dosing Regimen	Culture Type and Organism	24 hr %Time Above Minimum Inhibitory Concentration ^b
1	15	44.2	Septic shock	99	20mg/kg q8	Wound, MSSA	35%
2	33	27.7	Respiratory distress	133	20 mg/kg q8	Urine, <i>Enterococcus faecalis</i> , <i>Enterobacter aerogenes</i>	100%
3	13	41.9	Altered mental status, hematemesis	67	2 g q8	Culture negative	100%
4	4	15.6	Respiratory distress	156	20 mg/kg q8	Respiratory, MSSA	32%
5	19	68.5	Septic shock	117	20mg/kg q8	Culture negative	100%
6	0.6	5.7	Planned abdominal surgery	141	20mg/kg q8	Respiratory, <i>Pseudomonas aeruginosa</i>	80%
7	13	18.8	Hypoxia	213	40 mg/kg q8	Culture negative	83%
8	9	29	Septic shock	108	20mg/kg q8	Culture negative	31%
9	1	7.7	Respiratory failure	140	20mg/kg q8	Culture negative	28%
10	3	13.2	Septic shock	46	20 mg/kg q12	Culture negative	96%
11	2	16.2	Respiratory failure, septic shock	292	20mg/kg q8	Respiratory, <i>Haemophilus influenzae</i>	35%
12	17	39.9	Respiratory failure, septic shock	174	1 g q8	Urine, 3+ organisms not speciated	61%
13	0.4	4.5	Respiratory failure, seizure	208	40 mg/kg q8	Blood, <i>Escherichia coli</i>	63%
14	30	26	Respiratory distress	164	20mg/kg q8	Respiratory, <i>P. aeruginosa</i> , <i>Candida tropicalis</i>	74%
15	1.4	8.6	Unplanned abdominal surgery	108	20 mg/kg q8	Culture negative	65%
16	3	15	Respiratory distress	219	20mg/kg q8	Blood, <i>Klebsiella pneumoniae</i>	26%
17	2	9.8	Unplanned abdominal surgery, septic shock	57	20 mg/kg q8	Blood, <i>E. coli</i>	95%
18	4	12.9	Respiratory distress	134	20 mg/kg q8	Blood and respiratory, <i>Candida albicans</i>	91%
19	13	44.5	Sepsis	129	40 mg/kg q8	Culture negative	84%
20	14	25.5	Sepsis	233	20 mg/kg q8	Urine, <i>K. pneumoniae</i>	54%

(Continued)

TABLE 1. (Continued)

^aEstimated glomerular filtration rate, by Schwartz $k = 0.413$ for patients < 18 yr, by Chronic Kidney Disease Epidemiology Collaboration for patients ≥ 18 yr.

^bPercentage of first 24 hr where concentrations were above a minimum inhibitory concentration of 4 µg/mL.

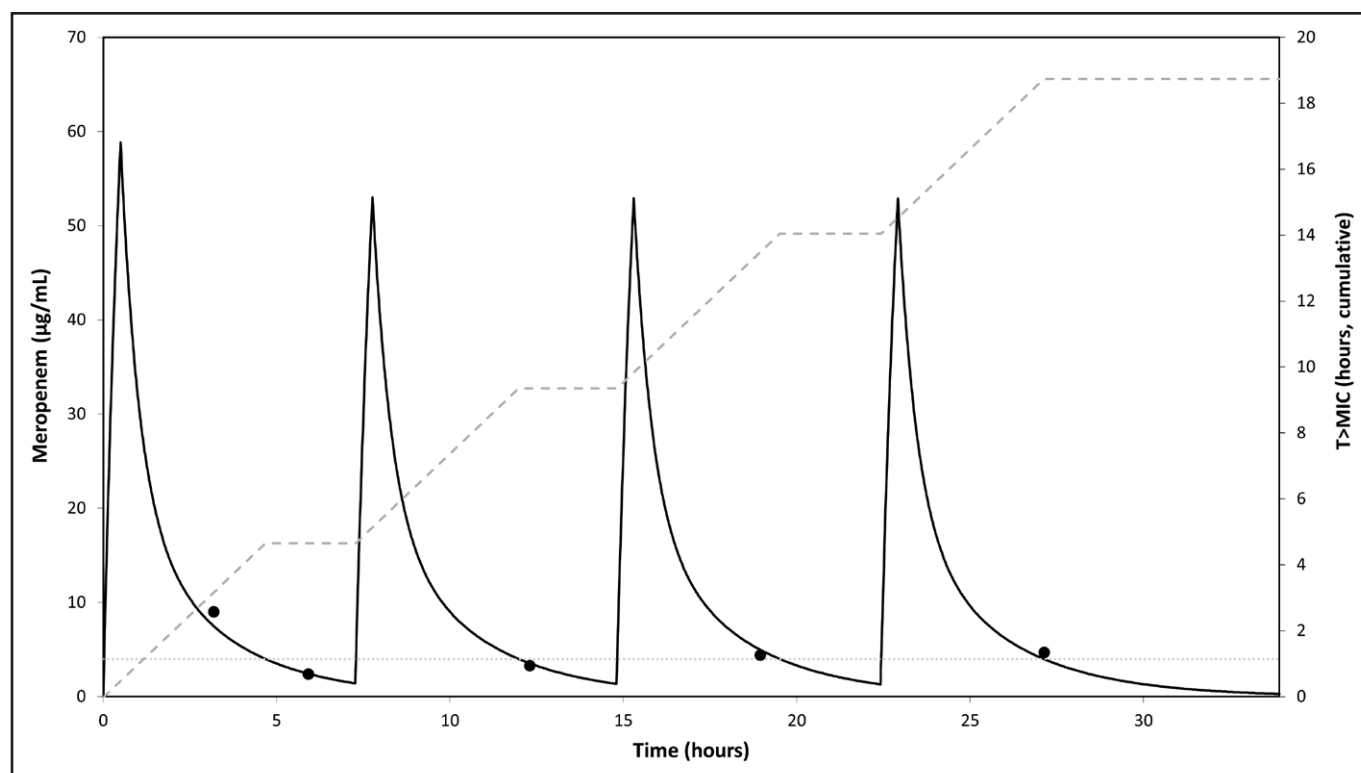


Figure 1. Example of a fitted concentration-time profile. The *black circles* represent measured opportunistic meropenem concentrations, and the *solid black line* reflects the fitted concentration-time profile based on patient characteristics, reference model, measured patient concentrations, and Bayesian estimation. The *gray dotted line* shows a minimum inhibitory concentration (MIC) of 4 µg/mL. The *gray dashed line* shows total cumulative time (hr) above MIC of 4 µg/mL.

The fitted concentration-time profile for each patient was visually inspected for quality and deemed adequate by our team. Residuals for each estimated value were compared against the observed values at those time points.

Finally, each patient was labeled as a “target attainer” or “non target attainer” for the $T > \text{MIC}$ threshold of interest ($40\%T > \text{MIC } 4 \mu\text{g/mL}$). Time between PICU admission and meropenem initiation was assessed to identify newly vs. ongoing critically ill patients. Patient characteristics, fluid status factors, pharmacokinetic parameters, and severe sepsis outcomes were compared between target attainers and non target attainers via chi-square, Fisher exact, or Mann-Whitney U tests. No corrections were made for multiple comparisons. Continuous variables were compared via Pearson correlation, with a p value of less than 0.05 considered significant.

RESULTS

Twenty-nine patients met inclusion criteria for the cohort. Admission diagnoses, meropenem dosing regimens, and culture data are summarized in Table 1. Four patients received CRRT. No patients received plasmapheresis. Fluid

balance data are shown in Table S2 (<http://links.lww.com/PCC/C541>). Only 16 of 29 patients had positive cultures, of which six of 16 had meropenem MIC data (Table S3, <http://links.lww.com/PCC/C541>). Median age and weight were 4 years and 16.3 kg, respectively (Table 2). Thirteen of 29 patients had a complicated course, with four deaths.

Meropenem Concentrations

Among the 29 patients, 80 concentrations were included in the analysis (median, 2 per patient [range, 1–7 per patient]). Three samples were below the LLQ and were imputed. Most opportunistic samples were from the mid-to-late dosing interval, that is, 2–6 hours post-dose (Fig. 2). No absolute residual was greater than 90% of the observed value, and the mean of residuals relative to observed values for each patient was between –61% and +38%.

Overall TA and Comparison of Target Attainers vs. Non Target Attainers

Twenty-one of 29 patients met the TA threshold of $40\%T > \text{MIC } 4 \mu\text{g/mL}$. On univariate analysis, without

Patient Characteristics, Therapies Required, Pharmacokinetic Parameters, and Outcomes for Severe Sepsis Patients That Attained the Meropenem Concentration Target of Time Above Minimum Inhibitory Concentration 4 µg/mL for At Least 40% of the First 24 Hours (*n* = 21), Compared With Those That Did Not Attain the Target (*n* = 6), and for Patients That Started Meropenem for Severe Sepsis Within 24 Hours of PICU Admission (*n* = 17) Compared With Those That Did Not (*n* = 12)

Variable	All Patients (<i>n</i> = 29)	Attained 40%T > MIC 4 µg/mL Over First 24 hr (<i>n</i> = 21)	Did Not Attain 40%T > MIC 4 µg/mL Over First 24 hr (<i>n</i> = 8)	Started Meropenem Within 24 hr (<i>n</i> = 17)	Started Meropenem After 24 hr (<i>n</i> = 12)
Patient characteristics					
Age, yr	4.0 (2–14.5)	8.0 (1.7–17)	3.5 (2.3–12.8)	8 (2–16)	3.5 (1.3–13)
Female sex, <i>n</i>	10	6	4	6	4
Weight, kg	16.3 (12.3–40.9)	18.8 (11.2–40.9)	15.9 (13.2–40.4)	25.5 (11.5–47.1)	14.4 (12.1–27.3)
Chronic kidney disease, <i>n</i>	4	3	1	2	2
Clinical status during meropenem					
eGFR, mL/min/1.73 m ²	133 (104–169)	129 (81–169)	143 (113–204)	122 (104–165)	133 (82–201)
eGFR > 130 mL/min/1.73 m ² , <i>n</i>	15	10	5	8	7
Required ≥ 40 mL/kg bolused fluid, <i>n</i>	11	8	3	10 ^a	1 ^a
Required positive pressure ventilation above baseline, <i>n</i>	22	18	4	12	10
Required vasopressors, <i>n</i>	13	11	2	7	6
On continuous renal replacement therapy, <i>n</i>	4	4	0	1	3
Fluid resuscitation and status					
Bolused fluid within 12 hr, mL/ admit-kg-bodyweight	17.2 (0–55.9)	16.6 (0–65.3)	24.4 (1.3–46.4)	46.0 (18.8–74.3) ^b	0 (0–8) ^b
Volume status at 24 hr, %-admit-bodyweight	4.8 (1.9–9.6)	5.0 (1.9–9.6)	3.9 (–0.1 to 8.9)	5.5 (2.8–10)	3.2 (0.6–6.1)
Pharmacodynamics					
Time concentration above MIC 4 µg/mL for first 24 hr, %	74 (36–95)	87 (68–98) ^b	31 (27–35) ^b	61 (33–94)	83 (73–99)
Achieved 40%T > MIC 4 µg/mL target, <i>n</i>	21	21 ^b	0 ^b	11	10

(Continued)

TABLE 2. (Continued)

Patient Characteristics, Therapies Required, Pharmacokinetic Parameters, and Outcomes for Severe Sepsis Patients That Attained the Meropenem Concentration Target of Time Above Minimum Inhibitory Concentration 4 µg/mL for At Least 40% of the First 24 Hours (n = 21), Compared With Those That Did Not Attain the Target (n = 6), and for Patients That Started Meropenem for Severe Sepsis Within 24 Hours of PICU Admission (n = 17) Compared With Those That Did Not (n = 12)

Variable	All Patients (n = 29)	Attained 40%T > MIC 4 µg/mL Over First 24 hr (n = 21)	Did Not Attain 40%T > MIC 4 µg/mL Over First 24 hr (n = 8)	Started Meropenem Within 24 hr (n = 17)	Started Meropenem After 24 hr (n = 12)
Pharmacokinetics					
Volume of distribution, L/70 kg	34 (26.3–41.4)	34.7 (26.8–42.8)	33.9 (22.5–36.9)	34.7 (23.8–41.2)	33.3 (26.8–42.5)
Clearance, L/hr/70 kg ^{0.75}	14.3 (8.5–22.8)	10.5 (8.1–15.4) ^b	27.4 (23.9–39.5) ^b	18.4 (8.5–25.3)	10.9 (8.4–19.5)
Meropenem start time to PICU admit, hr	15 (1–92)	20 (2–107)	6 (–2 to 42)	2 (0–14) ^b	107 (62–294) ^b
Severe sepsis course outcomes					
Positive cultures, n	16	11	5	9	7
ICU-free days at 28 d, d	15 (0–23)	15 (0–21)	19 (1–27)	21 (9–26) ^a	0 (0–16) ^a
Died by 28 d, n	4	4	0	1	3
Complicated course, n	13	11	2	5	8
Organ failures on day 1, organ systems	1 (0–2)	1 (0–2)	1 (1–1)	1 (0–2)	2 (0–2)

eGFR = estimated glomerular filtration rate, T > MIC = time above minimum inhibitory concentration.

^ap < 0.01.

^bp < 0.001.

Complicated course: Death by 28 d or persistence of 2+ organ failures at day 7. eGFR defined by Schwartz $k = 0.413$ for patients < 18 yr, by Chronic Kidney Disease Epidemiology Collaboration for patients ≥ 18 yr.

All values shown are counts or median with interquartile range.

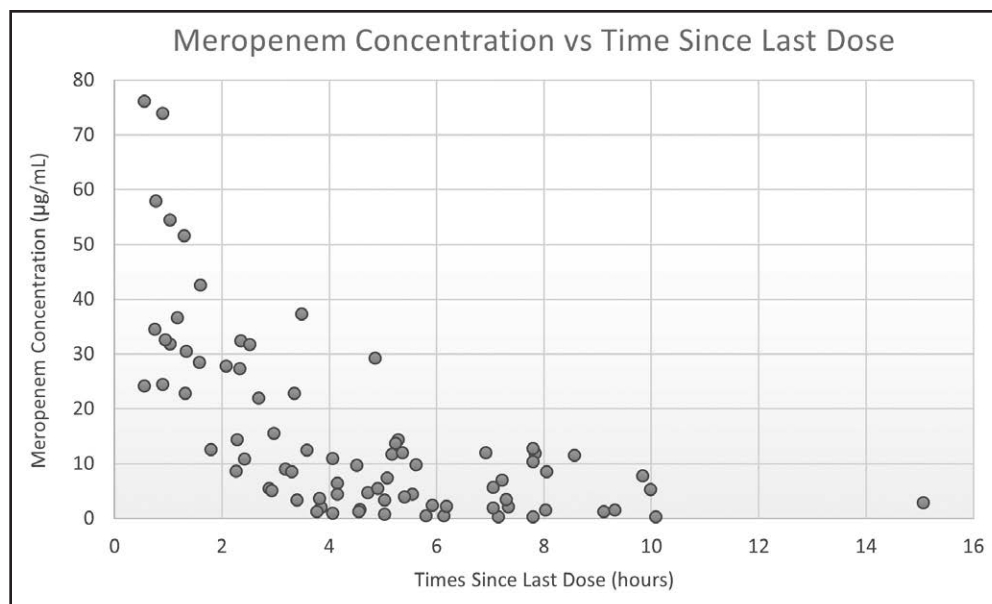


Figure 2. Concentrations vs. time since the last meropenem dose for opportunistic samples collected from all patients in the population (29 patients, 80 samples). There is significant variability in the peak values, that is, those falling 30–90 min after the start of infusion (which run over 30 min).

WHAT THIS STUDY MEANS

- In our cohort of 29 children with severe sepsis, eight did not achieve meropenem concentration targets chosen to treat potentially resistant organisms early in course. Higher meropenem clearance was associated with this outcome.
- Creatinine-estimated glomerular filtration rate (eGFR) may not be the best direct marker of meropenem clearance for making dosing decisions early in illness. However, eGFR was negatively correlated with overall $T > MIC$.
- Further investigation into risk factors for high meropenem clearance could identify septic children most likely to benefit from concentration monitoring and alternative empiric dosing.

correction for multiple comparisons, we failed to identify any associations between TA, vs. not, and patient characteristics, clinical status, or fluid resuscitation/balance status. The pharmacokinetic data showed an association between TA and median meropenem clearance with higher clearance in the non target attaining group (27.4 vs. 10.5 L/hr/70 kg^{0.75}; $p < 0.001$). Using Pearson correlation, clearance and %T > MIC had a correlation coefficient of -0.84 ($p < 0.001$). We failed

to identify any association between Vd and TA grouping or Vd and %T > MIC ($p = 0.35$).

We failed to identify an association with TA grouping and eGFR. However, eGFR was correlated with %T > MIC as continuous variable (correlation coefficient, -0.38 ; $p = 0.04$). We failed to find a correlation between eGFR and meropenem clearance at 24 hours ($p = 0.10$), between net fluid balance and Vd at 24 hours ($p = 0.97$), or between fluid bolus volume and Vd ($p = 0.39$).

TA and Comparison of Newly vs. Ongoing Critically Ill Patients

Of 29 patients, 17 were started on meropenem within 24 hours of PICU admission. Patient clinical and pharmacokinetic characteristics for this subgroup are shown in Table 2. Ten of 17 newly critically ill patients received greater than 40 mL/kg in bolus fluid compared with one of 12 patients with ongoing critical illness, with greater bolus fluid relative to admission bodyweight given to the newly critically ill group: median (interquartile range [IQR]) of 46 mL/kg (18.8–74.3 mL/kg) compared with 0 mL/kg (0–8 mL/kg; $p < 0.001$).

When comparing 40%T > MIC 4 µg/mL target attainers to non target attainers among the newly critically ill patients, 11 of 17 met this target (Table 3). There was an association between meropenem clearance and TA, with higher median clearance in non target attainers (27.4 L/hr/70 kg^{0.75} [IQR, 23.4–42.6 L/hr/70 kg^{0.75}] vs. 9.9 L/hr/70 kg^{0.75} [IQR, 8.1–21.3 L/hr/70 kg^{0.75}], with $p < 0.01$). We failed to identify notable association between other parameters (e.g., eGFR, fluid boluses, fluid balance, and Vd) and TA.

DISCUSSION

This pharmacokinetic study of meropenem shows that critically ill children with severe sepsis do not

TABLE 3.

Patient Characteristics, Therapies Required, Pharmacokinetic Parameters, and Outcomes for Severe Sepsis Patients That Attained the Meropenem Concentration Target of Time Above Minimum Inhibitory Concentration 4 µg/mL for At Least 40% of the First 24 Hours (*n* = 11), Compared With Those That Did Not Attain the Target (*n* = 6), for Only Those Patients Starting Meropenem Within 24 Hours of PICU Admission (*n* = 17)

Variable	Attained 40%T > MIC 4 µg/mL Across First 24 hr (<i>n</i> = 11)	Did Not Attain 40%T > MIC 4 µg/mL Across First 24 hr (<i>n</i> = 6)
Patient characteristics		
Age, yr	8 (2–17)	6.5 (2–14.3)
Female sex, <i>n</i>	4	2
Weight, kg	25.5 (9.8–49.9)	22.6 (13.6–50.7)
Chronic kidney disease, <i>n</i>	1	1
Clinical status during meropenem		
eGFR, mL/min/1.73 m ²	120 (95–174)	143 (106–190)
eGFR > 130 mL/min/1.73 m ² , <i>n</i>	4	4
Required ≥ 40 mL/kg bolused fluid, <i>n</i>	7	3
Required positive pressure ventilation above baseline, <i>n</i>	9	3
Required vasopressors, <i>n</i>	6	1
On continuous renal replacement therapy, <i>n</i>	1	0
Fluid resuscitation and status		
Bolused fluid within 12 hr, mL/admit-kg-bodyweight	63 (17–107)	34 (15–53)
Volume status at 24 hr, %-admit-bodyweight	7.5 (2.4–9.8)	5.1 (1.3–10.5)
Pharmacodynamics		
Time concentration above MIC 4 µg/mL for first 24 hr, %	87 (61–96) ^b	31 (27–35) ^b
Achieved 40%T > MIC 4 µg/mL target, <i>n</i>	11 ^b	0 ^b
Pharmacokinetics		
Volume of distribution, L/70 kg	37.5 (29.5–42.6)	29.8 (21–36.8)
Clearance, L/hr/70 kg ^{0.75}	9.9 (8.1–21.3) ^a	27.4 (23.4–42.6) ^a
Meropenem start time to PICU admit, hr	3 (0–15)	1 (–3 to 11)
Severe sepsis course outcomes		
Positive cultures, <i>n</i>	5	4
ICU-free days at 28 d, d	16 (0–26)	25 (13–27)
Died by 28 d, <i>n</i>	1	0
Complicated course, <i>n</i>	3	2
Organ failures on day 1, organ systems	1 (0–2)	1 (1–1)

eGFR = estimated glomerular filtration rate, T > MIC = time above minimum inhibitory concentration.

^a*p* < 0.01.

^b*p* < 0.001.

Complicated course: Death by 28 d or persistence of 2+ organ failures at day 7. eGFR defined by Schwartz *k* = 0.413 for patients < 18 yr, by Chronic Kidney Disease Epidemiology Collaboration for patients ≥ 18 yr.

All values shown are counts or median with interquartile range.

consistently meet TA thresholds suggested to treat antibiotic-resistant organisms within the first 24 hours of therapy. Twenty-one of 29 patients in our 2018–2021 cohort met a target of $40\%T > MIC$ $4 \mu\text{g/mL}$. Although there is lack of consensus on what empirical $\%T > MIC$ goals should be targeted for critically ill patients, there is agreement that early initiation of effective antibiotics saves lives. These findings suggest that merely infusing an antibiotic within a goal timeframe is not enough to yield effective therapy for all critically ill children—the right dose at the right time is needed.

Higher meropenem clearance was associated with failure to attain target concentrations. The median meropenem clearance in non target attaining patients from this study ($27.4 \text{ L/hr/70 kg}^{0.75}$; 95% CI, $23.8\text{--}41.3 \text{ L/hr/70 kg}^{0.75}$) appears similar to the Saito et al (18) population clearance of $25.4 \text{ L/hr/70 kg}^{0.75}$ (95% CI, $20.3\text{--}29.3 \text{ L/hr/70 kg}^{0.75}$) for patients with greater than or equal to 2 aspects of SIRS. Of note, the median eGFR of the Saito et al (18) population was 38.2 mL/min . Our clearance for non target attainers was higher than the $6.82 \text{ L/hr/70 kg}^{0.75}$ (normalized to eGFR of $100 \text{ mL/min/1.73 m}^2$) reported by Rapp et al (17) for PICU patients with high rates of renal dysfunction and CRRT. Interestingly, despite the non target attaining group having over twice the meropenem clearance compared with the target attaining group (which had median clearance $10.5 \text{ L/hr/70 kg}^{0.75}$; 95% CI, $8.5\text{--}14.3 \text{ L/hr/70 kg}^{0.75}$; thus, more similar to the group by Rapp et al [17]), we failed to identify the difference in day 1 median eGFR between the target attaining and non target attaining groups (129 vs. $143 \text{ mL/min/1.73 m}^2$; $p = 0.31$). eGFR and $\%T > MIC$, in addition to clearance and $\%T > MIC$, however, were correlated as continuous variables. Most meropenem is cleared unmetabolized via the kidneys; thus, renal function should reflect the majority of clearance (34). The failure to identify an association using the categorical renal findings likely reflects the fact that creatinine-informed eGFR formulas like the bedside Schwartz or CKD-EPI may not be accurate early measures of renal function in all critically ill children, both due to delayed laboratory evidence of kidney injury and low muscle mass typical of children with chronic illnesses. All four CRRT patients achieved the $40\%T > MIC$ target despite presumed circuit contributions to clearance, although it should be noted that the model by Saito et al (18) used in the analysis did not find CRRT to be a covariate on clearance.

We failed to find an association between meropenem Vd and TA grouping or between fluid resuscitation status and Vd. Meropenem Vd in these patients (34.7 L/70 kg ; 95% CI, $28.6\text{--}42 \text{ L/70 kg}$ vs. 33.9 L/70 kg ; 95% CI, $21.4\text{--}37 \text{ L/70 kg}$ for target attainers vs. non target attainers, respectively) were lower than the 40.6 L/70 kg reported by Rapp et al (17), but similar to the 33.6 L/70 kg (95% CI, $22.4\text{--}49 \text{ L/70 kg}$) reported by Saito et al (18) for the final model median Vd bootstrap. We failed to find correlation between Vd and net fluid balance or between Vd and fluid bolus volume across the cohort. While the addition of nonantibiotic containing IV fluids might be expected to expand Vd and “dilute” the intravascular space, higher Vd can extend half-life, and the confounding impact of capillary leak, dehydration, and vasoactive use may serve as balancing factors.

Patient outcomes did not differ between target attainers and non target attainers. We failed to identify any p value of less than 0.001 association between outcomes (including ICU-free days, complicated course, or death) and TA grouping. This may reflect limited power or small sample size, or the fact that few patients in this cohort grew resistant bacteria as the source of illness. A larger study may be needed to demonstrate associations between early TA and clinical outcomes in this population.

Variability in meropenem pharmacokinetics and early TA in critically ill patients reflect opportunity for personalized medicine. The variability in estimates of clearance and Vd, and subsequently in TA in our 2018–2021 cohort suggest that beta-lactam dosing based solely on weight and bedside serum creatinine-based eGFR may not be enough to ensure adequate TA in critically ill children. This interindividual pharmacokinetic variability might suggest need for therapeutic drug monitoring or model-informed precision dosing (MIPD), a technique that uses pharmacokinetic models and measured concentrations to inform dosing and achieve a predetermined therapeutic target (35–37). Although studies examining impact of MIPD on outcomes in adults have shown variable benefit (38–40), further studies could be used to identify specific subpopulations with pharmacokinetic parameter variability who might benefit from personalized antibiotic dosing.

The benefit of using PopPK models with Bayesian estimation for this study's analysis is that no assumptions are made about steady-state status thus allowing

for estimation of $T > MIC$ and pharmacokinetic parameters early in dosing course. Use of random opportunistic rather than midpoint or trough only sampling allows for capturing of dynamic PICU patient physiology. Limitations of this methodology relate to the number of available samples and uncertainties inherent to the applied model.

There were several limitations to the study. First, the sample size was small but similar to other studies of meropenem pharmacokinetics (15, 16, 18), and our cohort included only patients newly starting meropenem for severe sepsis. Second, use of opportunistic sampling resulted in limited samples around peaks (30–90 min after dose) or troughs (0–60 min before next dose) and uncertainty around subsequent estimates of V_d and clearance. Third, this study was not designed to consider meropenem losses from nonurine fluid outputs, which may be relevant in some patients. Finally, the PopPK model used in this analysis yielded variable fits for each patient but generally produced reasonably low mean bias. In sepsis, factors that affect meropenem pharmacokinetics can change rapidly and it can be difficult for any one model with set parameters to describe a patient across time.

CONCLUSIONS

Many patients do not meet stringent early TA goals for meropenem in severe sepsis. Characterization as a non target attainer is associated with higher clearance than target attainers. Continuous data show that early eGFR correlated with $T > MIC$. Further investigation into risk factors for high meropenem clearance could help ensure adequate early dosing regimens for septic children.

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